

# ANALOGUE FRONT ENDS For HEART RATE MONITORS

Analogue front ends (AFE) are a set of analogue signal conditioning circuitry that are used to interface sensors to an ADC or a microcontroller. In applications like heart rate monitors, AFEs bear tremendous importance. Let's find out why

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**H**earth rate is one of the critical vital signs in humans. For a healthy adult, it should be somewhere between 60 to 80 beats per minute. Like all other vital signs, the heart rate can change according to the person's location, physical and mental health, and even emotions. Naturally, the heart rate increases in response to an accident, injury, or trauma. Not just that, the rhythm, strength, and the time gap between the beats of your heart can also fluctuate.

The monitoring and tracking of heart rate is a relatively simple task for modern electronics. Moreover, keeping track of your heart rate can be very helpful, not just for medical precautionary purposes but also for developing a fitness regime. That's why modern wearable devices like smartwatches and activity trackers widely utilise the heart rate monitoring function. It is especially useful in determining whether one is exercising at the correct intensity or not.

In addition, such devices can detect possible diseases like tachycardia (fast heart rate), bradycardia (slow heart rate), arrhythmia (often changed heart rate), and others. Also, heart rate monitors are widely used in sports. With the help of a heart rate measurement, ath-

## Fitness tracking and history of heart rate monitoring

It was in 1977 that Polar Electro, a Finnish company, invented the first wireless electrocardiogram (ECG) heart rate monitor. It was meant to be a training aid for the Finnish National Cross Country Ski team. Soon, people realised that monitoring vital signs like heart rate and other exercise metrics are best left to technology, and heart rate monitors became popular. In 1984, Polar Electro launched the PE3000, a watch-like heart rate monitor with an integrated display. Today, heart rate monitoring is common and nearly everyone owns some sort of fitness tracking device.

letes can define appropriate load and rest intervals.

## Why heart rate monitors need AFEs

Generally, heart rate monitors employ sensors that are analogue in nature. The contractions and relaxations of the heart while it pumps blood can be viewed as electrical activities and hence, the signals or pulses are shown as waveforms. However, the controllers employed in a heart rate monitor can only interpret binary data, that is, digital signals. Hence, an analogue-to-digital converter (ADC) is needed. Moreover, the signal needs to be amplified and filtered to remove noise.

This is why AFEs are so important. They contain all the necessary components and circuitry to perform signal conditioning on the signals that come right from the sensors. In general, AFEs include analogue amplifiers, operational amplifiers, filters, sample and hold circuits, and



PE3000 by Polar Electro, one of the first wearable heart rate monitors with an integrated computer interface (Credit: [www.healthchecksystems.com](http://www.healthchecksystems.com))